

Q) Secure VPC Architecture with Bastion Host

Create a VPC named **Lab_VPC_<PRN>** with CIDR block 10.0.0.0/16. Create one Public Subnet (10.0.1.0/24) and one Private Subnet (10.0.2.0/24). Attach an Internet Gateway and configure route tables such that only the Public Subnet has internet access. Launch one EC2 instance in each subnet. Configure Security Groups so that the Public EC2 can be accessed from your local machine through SSH, while the Private EC2 can only be accessed from the Public EC2. Create a file containing your PRN number on the Public EC2, transfer it securely to the Private EC2, and verify the contents. Demonstrate successful SSH connectivity from Local Machine → Public EC2 → Private EC2.

Q) Secure VPC Architecture with Bastion Host

Step 1: Create a Custom VPC

A VPC (Virtual Private Cloud) provides an isolated network environment in AWS. We create a custom VPC to control networking, routing, and security.

UI Steps

1. Login to AWS Console.
2. Search **VPC**.
3. Click **Your VPCs**.
4. Click **Create VPC**.
5. Select **VPC only**.

Configuration

Field	Value
Name Tag	Lab_VPC_<prn>
IPv4 CIDR	10.0.0.0/16
IPv6 CIDR	No IPv6
Tenancy	Default

6. Click **Create VPC**.

Explanation of Attributes

- **10.0.0.0/16** → Provides 65,536 IP addresses.
- **Default Tenancy** → Uses shared AWS hardware.

Your VPCs

VPCsVPC encryption controls

Your VPCs (1/2) Info

Last updated less than a minute ago

Actions

Create VPC

Find VPCs by attribute or tag

	Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public...	IPv4 CIDR	
☐	Default	vpc-0a89913c46400da0f	Available	-	-	Off	172.31.0.0/16	-
☒	LAB_VPC_260240123060	vpc-0b47b98149ec8fc11	Available	-	-	Off	10.0.0.0/16	-

VPC > Your VPCs > vpc-0b47b98149ec8fc11

Actions

VPC dashboard

AWS Global View

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Network ACLs

Security groups

vpc-0b47b98149ec8fc11 / LAB_VPC_260240123060

Actions

Details Info

VPC ID

vpc-0b47b98149ec8fc11

DNS resolution

Enabled

Main network ACL

acl-03a5fe925eaa4286a

IPv6 CIDR (Network border group)

-

Encryption control ID

-

State

Available

Tenancy

default

Default VPC

No

Network Address Usage metrics

Disabled

Encryption control mode

-

Block Public Access

Off

DHCP option set

dopt-0eaaf03657f55f9c3

IPv4 CIDR

10.0.0.0/16

Route 53 Resolver DNS Firewall rule groups

-

DNS hostnames

Disabled

Main route table

rtb-0d1e295bcb91537f6

IPv6 pool

-

Owner ID

561565006234

Resource map

CIDRs

Flow logs

Tags

Integrations

Resource map Info

Show all details

VPC

Your AWS virtual network

Subnets (0)

Subnets within this VPC

Route tables (1)

Route network traffic to resources

Step 2: Create Public Subnet

The Public Subnet will host the Bastion Host (Public EC2) which is accessible from the internet.

UI Steps

1. Open **Subnets**.
2. Click **Create Subnet**.

Configuration

Field	Value
VPC	Lab_VPC_<prn>
Name	Public_Subnet
AZ	ap-south-1a
CIDR	10.0.1.0/24

3. Click **Create Subnet**.

Explanation of Attributes

- **10.0.1.0/24** → Provides 256 IP addresses.
- Public subnet will receive internet access through IGW.

You have successfully created 1 subnet: subnet-03b2efebd611ba58f

Subnets (1/4) Info

Last updated less than a minute ago

Actions

Create subnet

Find subnets by attribute or tag

	Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR	IPv6 CIDR
☐	default 1	subnet-01301be8126ae9e2b	Available	vpc-0a89913c46400da0f Defa...	Off	172.31.32.0/20	-
☐	default 2	subnet-0734393e5c2afccaf	Available	vpc-0a89913c46400da0f Defa...	Off	172.31.0.0/20	-
☐	default 3	subnet-06cbbd81834322721	Available	vpc-0a89913c46400da0f Defa...	Off	172.31.16.0/20	-
☒	Public_Subnet	subnet-03b2efebd611ba58f	Available	vpc-0b47b98149ec8fc11 LAB_...	Off	10.0.1.0/24	-

Details

Subnet ID

 subnet-03b2efebd611ba58f

IPv4 CIDR

 10.0.1.0/24

Availability Zone

 ap-s1-az1 (ap-south-1a)

Network ACL

-

Auto-assign customer-owned IPv4 address

No

IPv6 CIDR reservations

-

Resource name DNS AAAA record

Disabled

Subnet ARN

 arn:aws:ec2:ap-south-1:561565006234:subnet/subnet-03b2efebd611ba58f

Available IPv4 addresses

 251

Network border group

 ap-south-1

Default subnet

No

Customer-owned IPv4 pool

-

IPv6-only

No

DNS64

Disabled

State

 Available

IPv6 CIDR

-

VPC

[vpc-0b47b98149ec8fc11 |](#)
[LAB_VPC_260240123060](#)

Auto-assign public IPv4 address

No

Outpost ID

-

Hostname type

IP name

Owner

 561565006234

Block Public Access

 Off

IPv6 CIDR association ID

-

Route table

-

Auto-assign IPv6 address

No

IPv4 CIDR reservations

-

Resource name DNS A record

Disabled

Step 3: Create Private Subnet

The Private Subnet hosts internal resources which should not be directly accessible from the internet.

UI Steps

1. Click **Create Subnet**.

Configuration

Field	Value
VPC	Lab_VPC_<prn>
Name	Private_Subnet
AZ	ap-south-1b
CIDR	10.0.2.0/24

2. Click **Create Subnet**.

Explanation of Attributes

- **10.0.2.0/24** → Private network segment.
- No direct internet access.

You have successfully created 1 subnet: subnet-064ec056db4ce6a21

Subnets (1/5) Info

Last updated less than a minute ago

Find subnets by attribute or tag

☐	Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR	IPv6 CIDR
☐	default 1	subnet-01301be8126ae9e2b	Available	vpc-0a89913c46400da0f Defa...	Off	172.31.32.0/20	-
☐	default 2	subnet-0734393e5c2afccaf	Available	vpc-0a89913c46400da0f Defa...	Off	172.31.0.0/20	-
☐	default 3	subnet-06cbbd81834322721	Available	vpc-0a89913c46400da0f Defa...	Off	172.31.16.0/20	-
☐	Public_Subnet	subnet-03b2efebd611ba58f	Available	vpc-0b47b98149ec8fc11 LAB_...	Off	10.0.1.0/24	-
☒	Private_Subnet	subnet-064ec056db4ce6a21	Available	vpc-0b47b98149ec8fc11 LAB_...	Off	10.0.2.0/24	-

subnet-064ec056db4ce6a21 / Private_Subnet

Actions

Details	
Subnet ID subnet-064ec056db4ce6a21	Subnet ARN arn:aws:ec2:ap-south-1:561565006234:subnet/subnet-064ec056db4ce6a21
IPv4 CIDR 10.0.2.0/24	Available IPv4 addresses 251
Availability Zone ap-s1-az3 (ap-south-1b)	Network border group ap-south-1
Network ACL -	Default subnet No
Auto-assign customer-owned IPv4 address No	Customer-owned IPv4 pool -
IPv6 CIDR reservations -	IPv6-only No
Resource name DNS AAAA record Disabled	DNS64 Disabled
State Available	IPv6 CIDR -
VPC vpc-0b47b98149ec8fc11 LAB_VPC_260240123060	Auto-assign public IPv4 address No
Outpost ID -	Hostname type IP name
Owner 561565006234	Block Public Access Off
IPv6 CIDR association ID -	Route table -
Auto-assign IPv6 address No	IPv4 CIDR reservations -
Resource name DNS A record Disabled	

Step 4: Create Internet Gateway

Internet Gateway allows internet communication for resources inside the Public Subnet.

UI Steps

1. Open **Internet Gateways**.
2. Click **Create Internet Gateway**.

Configuration

Field Value

Name Lab_IGW

3. Click **Create**.
4. Select IGW.
5. Click **Attach to VPC**.
6. Select **Lab_VPC_**.

Explanation of Attributes

- IGW enables internet routing.

Find internet gateways by attribute or tag					
☐	Name	Internet gateway ID	State	VPC ID	Owner
☐	default	igw-0923bd814e2cbffa3	Attached	vpc-0a89913c46400da0f Default	561565006234
☒	Lab_IGW	igw-067a71112cc1c2d38	Attached	vpc-0b47b98149ec8fc11 LAB_VPC_26...	561565006234

Step 5: Create Public Route Table

Route table defines how network traffic travels.

UI Steps

1. Open **Route Tables**.
2. Click **Create Route Table**.

Configuration

Field **Value**

Name Public_RT

VPC Lab_VPC_

3. Create Route Table.

Add Route

sdgWe are adding a route that allows instances in the **Public Subnet** to access the Internet through the **Internet Gateway (IGW)**.

UI Steps

1. Open **VPC Dashboard**.
2. Click **Route Tables** from the left menu.
3. Select **Public_RT**.
4. Click the **Routes** tab.
5. Click **Edit Routes**.
6. Click **Add Route**.

Fill the details:

Field	Value
Destination	0.0.0.0/0
Target	Internet Gateway

- 7. Click the **Target** dropdown.
- 8. Select your Internet Gateway (**Lab_IGW**).
- 9. Click **Save Changes**.

Explanation of Attributes

- **0.0.0.0/0** → Represents all IPv4 addresses (entire Internet).
- **Internet Gateway (IGW)** → Allows traffic to enter and leave the VPC through the Internet.

rtb-00132a17dad750b08 / Public_RT

Actions

Details

Route table ID
rtb-00132a17dad750b08

VPC
vpc-0b47b98149ec8fc11 | LAB_VPC_260240123060

Main
No

Owner ID
561565006234

Explicit subnet associations
subnet-03b2efebd611ba58f / Public_Subnet

Edge associations
-

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (2)

Both

Edit routes

Filter routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-067a71112cc1c2d38	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

Associate Public Route Table with Public Subnet

What are we doing and why?

A route table becomes effective only after it is associated with a subnet. Associating **Public_RT** with **Public_Subnet** makes that subnet internet accessible.

UI Steps

1. Select **Public_RT**.
2. Click the **Subnet Associations** tab.
3. Click **Edit Subnet Associations**.
4. Select:Public_Subnet (10.0.1.0/24)
5. Click **Save Associations**.

Verification

After saving, you should see:

Route Table: Public_RT

Associated Subnets:

✓ Public_Subnet

VPC dashboard < rtb-00132a17dad750b08 / Public_RT Actions

Details info

Route table ID: rtb-00132a17dad750b08

Main: No

Explicit subnet associations: subnet-03b2efebd611ba58f / Public_Subnet

Edge associations: -

VPC: vpc-0b47b98149ec8fc11 | LAB_VPC_260240123060

Owner ID: 561565006234

Routes Subnet associations Edge associations Route propagation Tags

Explicit subnet associations (1)

Find subnet association

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
Public_Subnet	subnet-03b2efebd611ba58f	10.0.1.0/24	-

Edit subnet associations

Subnets without explicit associations (1)

The following subnets have not been explicitly associated with any route tables and are therefore associated with the main route table:

Find subnet association

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
Private_Subnet	subnet-064ec056db4ce6a21	10.0.2.0/24	-

Edit subnet associations

Step 6: Create Private Route Table

Private subnet should remain isolated from internet.

UI Steps

- ## 1. Create another Route Table.

Configuration

Field	Value
Name	Private_RT
VPC	Lab_VPC_<prn>

Associate

Associate only with:

- Private_Subnet

Explanation of Attributes

- No internet route added.
- Subnet remains private.

VPC > Route tables

VPC dashboard

AWS Global View

Filter by VPC

Virtual private cloud

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Subnets

Route tables

Internet gateways

Egress-only internet gateways

Route tables (1/4)

Find route tables by attribute or tag

	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC	Owner ID
☐	default	rtb-062ae9de6234da21d	-	-	Yes	vpc-0a89913c4640da0f Defa...	561565006234
☐	-	rtb-0d1ec295bcb91537f6	-	-	Yes	vpc-0b47b98149ec8fc11 LAB...	561565006234
☒	Public_RT	rtb-00132a17dad750b08	subnet-03b2efebd511ba...	-	No	vpc-0b47b98149ec8fc11 LAB...	561565006234
☐	Private_RT	rtb-0bd6d6c49117c359f	subnet-064ec056db4ce6...	-	No	vpc-0b47b98149ec8fc11 LAB...	561565006234

VPC > Route tables > rtb-0bd6d6c49117c359f

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Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

rtb-0bd6d6c49117c359f / Private_RT

You have successfully updated subnet associations for rtb-0bd6d6c49117c359f / Private_RT.

Actions

Details

Route table ID

rtb-0bd6d6c49117c359f

Main

No

Explicit subnet associations

subnet-064ec056db4ce6a21 / Private_Subnet

Edge associations

-

VPC

vpc-0b47b98149ec8fc11 | LAB_VPC_260240123060

Owner ID

561565006234

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (1)

Filter routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	Create Route Table

Step 7: Enable Auto Assign Public IP

Public EC2 requires a public IP for SSH access.

UI Steps

1. Open Subnets.
2. Select Public_Subnet.
3. Edit Subnet Settings.
4. Enable:

Auto-assign Public IPv4 Address

5. Save.

VPC > Subnets > subnet-03b2efebd611ba58f > Edit subnet settings

ⓘ

Edit subnet settings

Info

Subnet

Subnet ID

Subnet Name

subnet-03b2efebd611ba58f

Public_Subnet

Auto-assign IP settings

Info

Enable AWS to automatically assign a public IPv4 or IPv6 address to a new primary network interface for an instance in this subnet.

☒

Enable auto-assign public IPv4 address

Info

☐

Enable auto-assign customer-owned IPv4 address

Info

Option disabled because no customer owned pools found.

Resource-based name (RBN) settings

Info

Specify the hostname type for EC2 instances in this subnet and optional RBN DNS query settings.

☐

Enable resource name DNS A record on launch

Info

☐

Enable resource name DNS AAAA record on launch

Info

Hostname type

Info

☐ Resource name

☒ IP name

Step 8: Create Security Group for Public EC2

Allows SSH from your laptop to Bastion Host.

UI Steps

Create Security Group.

Configuration

Field	Value
Name	Public_SG
VPC	Lab_VPC_<prn>

Inbound Rules

Type	Port	Source
SSH	22	My IP

Explanation

- Only your machine can SSH.

VPC > Security Groups > Create security group

Create security group info

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name info

Public_SG

Name cannot be edited after creation.

Description info

Allows SSH access to developers

VPC info

vpc-0b47b98149ec8fc11 (LAB_VPC_260240123060)

Inbound rules info

Type info

SSH

Protocol info

TCP

Port range info

22

Source info

Anywhere...

Description - optional info

0.0.0.0/0

Add rule

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Outbound rules info

Type info

All traffic

Protocol info

All

Port range info

All

Destination info

Custom

Description - optional info

0.0.0.0/0

Delete

Security Groups (1/6) Info

Find security groups by attribute or tag

Actions

Export security groups to CSV

Create security group

	Name	Security group ID	Security group name	VPC ID	Description	Owner	Int
☐	-	sg-0b26d1ce8599ef088	launch-wizard-1	vpc-0a89913c46400da0f	launch-wizard-1 created 2026-05-07T0...	561565006234	3 F
☐	-	sg-046b599b45e23ee09	launch-wizard-3	vpc-0a89913c46400da0f	launch-wizard-3 created 2026-05-07T0...	561565006234	1 F
☒	-	sg-0e0b9a957b1563f24	Public_SG	vpc-0b47b98149ec8fc11	Allow ssh pub	561565006234	1 F
☐	-	sg-047a3c61012a85edf	default	vpc-0b47b98149ec8fc11	default VPC security group	561565006234	1 F
☐	-	sg-0d54cd8f49c2706e0	default	vpc-0a89913c46400da0f	default VPC security group	561565006234	1 F
☐	-	sg-070a5885cbc6bbbff9	launch-wizard-2	vpc-0a89913c46400da0f	launch-wizard-2 created 2026-05-07T0...	561565006234	1 F

sg-0e0b9a957b1563f24 - Public_SG

DetailsInbound rulesOutbound rulesSharingVPC associationsRelated resources - newTags

Details

Security group namePublic_SG

Owner561565006234

Security group IDsg-0e0b9a957b1563f24

Inbound rules count1 Permission entry

DescriptionAllow ssh pub

Outbound rules count1 Permission entry

VPC IDvpc-0b47b98149ec8fc11

sg-0e0b9a957b1563f24 - Public_SG

Actions

Details

Security group namePublic_SG

Owner561565006234

Security group IDsg-0e0b9a957b1563f24

Inbound rules count1 Permission entry

DescriptionAllow ssh pub

Outbound rules count1 Permission entry

VPC IDvpc-0b47b98149ec8fc11

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Inbound rules (1)

Search

Manage tags

Edit inbound rules

	Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
☐	-	sgr-0580a2a14eb894f21	IPv4	SSH	TCP	22	0.0.0.0/0	-

Step 9: Create Security Group for Private EC2

Allows SSH only from Bastion Host.

UI Steps

Create Security Group.

Configuration

Field	Value
Name	Private_SG

Inbound Rules

Type	Port	Source
SSH	22	Public_SG

Explanation

- Only Public EC2 can access Private EC2.

VPC

Security Groups

Create security group

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name

Private_SG

Name cannot be edited after creation.

Description

Private_SG

VPC

vpc-0b47b98149ec8fc11 (LAB_VPC_260240123060)

Inbound rules

Type

SSH

Protocol

TCP

Port range

22

Source

Custom

Q

sg-0e0b9a957b1563f24

Description - optional

Add rule

Outbound rules

Type

All traffic

Protocol

All

Port range

All

Destination

Custom

Q

0.0.0.0/0

Description - optional

Add rule

VPC > Security Groups

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Peering connections

Route servers

Security

Network ACLs

Security groups

PrivateLink and Lattice

Security Groups (1/7)

Find security groups by attribute or tag

Name	Security group ID	Security group name	VPC ID	Description	Owner	Int
-	sg-0b26d1ce8599ae088	launch-wizard-1	yvc-0a89913c46400da0f	launch-wizard-1 created 2026-05-07T0...	561565006234	3 F
-	sg-046b599b45e23ee09	launch-wizard-3	yvc-0a89913c46400da0f	launch-wizard-3 created 2026-05-07T0...	561565006234	1 F
-	sg-0e0b9a957b1563f24	Public_SG	yvc-0b47b98149ec8fc11	Allow ssh pub	561565006234	1 F
-	sg-047a3c61012d85edf	default	yvc-0b47b98149ec8fc11	default VPC security group	561565006234	1 F
-	sg-0a34c4d454f8636e8	Private_SG	yvc-0b47b98149ec8fc11	Private_SG	561565006234	1 F
-	sg-0d54c8f49c2706e0	default	yvc-0a89913c46400da0f	default VPC security group	561565006234	1 F

sg-0a34c4d454f8636e8 - Private_SG

Details

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Inbound rules (1)

Search

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
-	sgr-00fbc998cc407225f	-	SSH	TCP	22	sg-0e0b9a957b1563f2...	-

BWS

VPC > Security Groups

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PrivateLink and Lattice

Security Groups (1/7)

Find security groups by attribute or tag

Name	Security group ID	Security group name	VPC ID	Description	Owner	Int
-	sg-0b26d1ce8599ae088	launch-wizard-1	yvc-0a89913c46400da0f	launch-wizard-1 created 2026-05-07T0...	561565006234	3 F
-	sg-046b599b45e23ee09	launch-wizard-3	yvc-0a89913c46400da0f	launch-wizard-3 created 2026-05-07T0...	561565006234	1 F
-	sg-0e0b9a957b1563f24	Public_SG	yvc-0b47b98149ec8fc11	Allow ssh pub	561565006234	1 F
-	sg-047a3c61012d85edf	default	yvc-0b47b98149ec8fc11	default VPC security group	561565006234	1 F
-	sg-0a34c4d454f8636e8	Private_SG	yvc-0b47b98149ec8fc11	Private_SG	561565006234	1 F
-	sg-0d54c8f49c2706e0	default	yvc-0a89913c46400da0f	default VPC security group	561565006234	1 F

sg-0a34c4d454f8636e8 - Private_SG

Details

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Details

Security group name

Private_SG

Owner

561565006234

Security group ID

sg-0a34c4d454f8636e8

Inbound rules count

1 Permission entry

Description

Private_SG

Outbound rules count

1 Permission entry

VPC ID

yvc-0b47b98149ec8fc11

Step 10: Launch Public EC2 (Bastion Host)

Acts as a jump server to access private resources.

UI Steps

Launch Instance.

Configuration

Field	Value
Name	Public_EC2
AMI	Ubuntu
Type	t2.micro
Key Pair	Existing/New
VPC	Lab_VPC_<prn>
Subnet	Public_Subnet
Security Group	Public_SG
Public IP	Enable

Launch instance.

Instance summary for i-0acb7e10296dc0d55 (Public_EC2) Info

Updated less than a minute ago

Instance ID

i-0acb7e10296dc0d55

IPv6 address

-

Hostname type

IP name: ip-10-0-1-89.ap-south-1.compute.internal

Answer private resource DNS name

-

Auto-assigned IP address

13.206.96.82 [Public IP]

IAM role

-

IMDSv2

Required

Operator

-

Public IPv4 address

13.206.96.82 | [open address](#)

Instance state

Running

Private IP DNS name (IPv4 only)

ip-10-0-1-89.ap-south-1.compute.internal

Instance type

t3.micro

VPC ID

vpc-0b47b98149ec8fc11 (LAB_VPC_260240123060) [↗](#)

Subnet ID

subnet-03b2efebd611ba58f (Public_Subnet) [↗](#)

Instance ARN

arn:aws:ec2:ap-south-1:561565006234:instance/i-0acb7e10296dc0d55

Private IPv4 addresses

10.0.1.89

Public DNS

-

Elastic IP addresses

-

AWS Compute Optimizer finding

[Opt-in to AWS Compute Optimizer for recommendations.](#) | [Learn more](#) [↗](#)

Auto Scaling Group name

-

Managed

false

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

▼ Instance details Info

AMI ID

ami-07a00cf47dbbc844c

AMI name

ubuntu/images/hvm-ssd-gp3/ubuntu-resolute-26.04-amd64-server-20260421

Monitoring

disabled

Allowed image

-

Platform details

Linux/UNIX

Termination protection

Disabled

EC2

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Instances

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Save filter sets

Find instance by attribute or tag (case-sensitive)

1

Public_EC2

I-0acb7e10296dc0d55

Running

t3.micro

Initializing

ap-south-1a

Public IPv4 DNS

Public IPv4 ...

13.206.96.82

Elastic IP

IPv6 IPs

Monitoring

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

Instance summary

Instance ID

I-0acb7e10296dc0d55

IPv6 address

-

Hostname type

IP name: ip-10-0-1-89.ap-south-1.compute.internal

Answer private resource DNS name

-

Auto-assigned IP address

13.206.96.82 [Public IP]

IAM role

-

IMDSv2

Required

Public IPv4 address

13.206.96.82 | open address

Instance state

Running

Private IP DNS name (IPv4 only)

ip-10-0-1-89.ap-south-1.compute.internal

Instance type

t3.micro

VPC ID

vpc-0b47b08149ec8fc11 (LAB_VPC_260240123060)

Subnet ID

subnet-03b2efebd611ba58f (Public_Subnet)

Instance ARN

arn:aws:ec2:ap-south-1:561565006234:instance/I-0acb7e10296dc0d55

Private IPv4 addresses

10.0.1.89

Public DNS

-

Elastic IP addresses

-

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendations. | Learn more

Auto Scaling Group name

-

Managed

false

Step 11: Launch Private EC2

This server remains inaccessible directly from the internet.

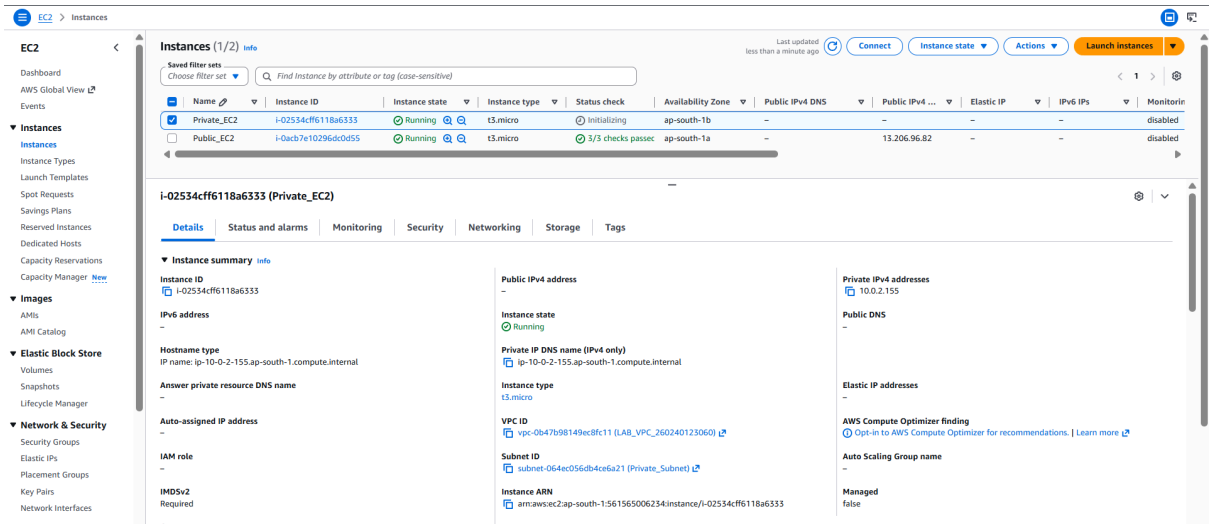
UI Steps

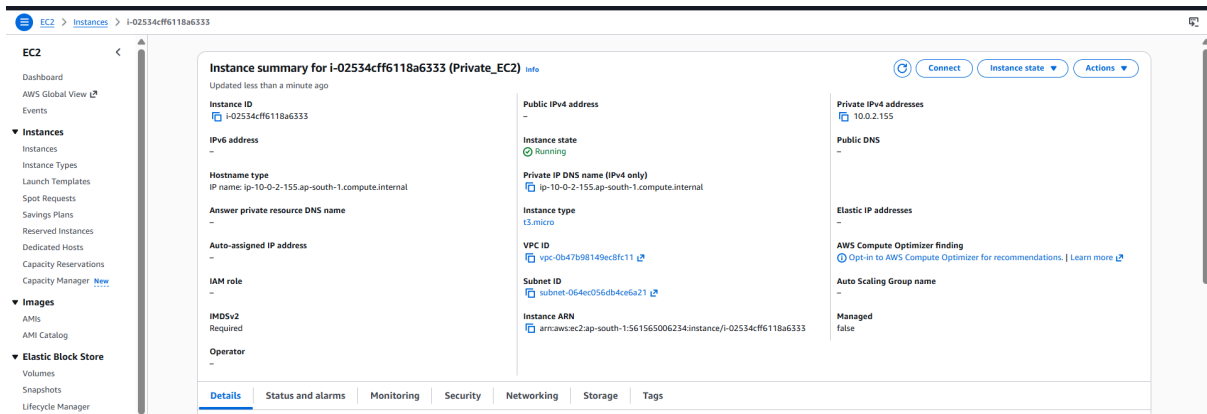
Launch Instance.

Configuration

Field	Value
Name	Private_EC2
AMI	Ubuntu
Type	t2.micro
VPC	Lab_VPC_<prn>
Subnet	Private_Subnet
Security Group	Private_SG
Public IP	Disable

Launch instance.





Step 12: Connect Local Machine to Public EC2

Verify Bastion Host accessibility.

Command

```
ssh -i lab.pem ubuntu@<PUBLIC_EC2_PUBLIC_IP>
```

Explanation

- `-i` → Private key.
- `ubuntu` → Ubuntu username.

Navigate to the folder containing the key:

```
cd C:\key
```

Remove inherited permissions:

```
icacls <<your_key_name>>.pem /inheritance:r
```

Grant permission only to your account:

```
icacls <<your_key_name>>.pem /grant:r "$($env:USERNAME):(R)"
```

Verify:

```
icacls <<your_key_name>>.pem
```

```
PS C:\key> icacls del_20260607_lab_Vpc.pem /inheritance:r
processed file: del_20260607_lab_Vpc.pem
Successfully processed 1 files; Failed processing 0 files
PS C:\key> icacls del_20260607_lab_Vpc.pem /inheritance:r
processed file: del_20260607_lab_Vpc.pem
Successfully processed 1 files; Failed processing 0 files
PS C:\key> icacls del_20260607_lab_Vpc.pem /grant:r "$($env:USERNAME):(R)"
processed file: del_20260607_lab_Vpc.pem
Successfully processed 1 files; Failed processing 0 files
PS C:\key> icacls del_20260607_lab_Vpc.pem
del_20260607_lab_Vpc.pem THEMIST\Lenovo:(R)
```

```
PS C:\key> ssh -i "del_20260607_lab_Vpc.pem" ubuntu@13.206.96.82
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1004-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sun Jun  7 17:10:44 UTC 2026

System load:  0.0           Temperature:   -273.1 C
Usage of /:   30.4% of 6.61GB Processes:      116
Memory usage: 26%          Users logged in: 0
Swap usage:   0%           IPv4 address for ens5: 10.0.1.89

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

ubuntu@ip-10-0-1-89:~$
```

Step 13: Copy Key to Public EC2

Private EC2 will be accessed from Public EC2.

Command (Local Machine)

```
scp -i lab.pem lab.pem ubuntu@<PUBLIC_EC2_PUBLIC_IP>:/home/ubuntu/
```

Explanation

- Securely transfers PEM key.

```
ubuntu@ip-10-0-1-89:~$ ls
del_20260607_lab_Vpc.pem
ubuntu@ip-10-0-1-89:~$ scp -i del_20260607_lab_Vpc.pem del_20260607_lab_Vpc.pem ubuntu@13.206.96.82:/home/ubuntu/
The authenticity of host '13.206.96.82 (13.206.96.82)' can't be established.
ED25519 key fingerprint is: SHA256:s0JLByQuixGivzJqSFPimAXMTVU4LsRl6oXGsYSYn4I
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added '13.206.96.82' (ED25519) to the list of known hosts.
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
@                WARNING: UNPROTECTED PRIVATE KEY FILE!          @
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
Permissions 0664 for 'del_20260607_lab_Vpc.pem' are too open.
It is required that your private key files are NOT accessible by others.
This private key will be ignored.
Load key "del_20260607_lab_Vpc.pem": bad permissions
ubuntu@13.206.96.82: Permission denied (publickey).
scp: Connection closed
```

Step 14: Create PRN File on Public EC2

Create a file that will be transferred to Private EC2.

Command

```
echo "PRN: 260240123060" > prn.txt
```

```
ubuntu@ip-10-0-1-89:~$ echo "PRN: 260240123060" > prn.txt
ubuntu@ip-10-0-1-89:~$ ls
del_20260607_lab_Vpc.pem  prn.txt
```

Step 15: SSH from Public EC2 to Private EC2

Verify Bastion Host architecture.

Command

```
chmod 400 lab.pem
```

```
ssh -i lab.pem ubuntu@<PRIVATE_EC2_PRIVATE_IP>
```

```
ubuntu@ip-10-0-1-89:~$ chmod 400 del_20260607_lab_Vpc.pem
ubuntu@ip-10-0-1-89:~$ ssh -i del_20260607_lab_Vpc.pem ubuntu@10.0.2.155
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1004-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sun Jun  7 17:27:58 UTC 2026

System load:  0.0           Temperature:   -273.1 C
Usage of /:   30.4% of 6.61GB Processes:      115
Memory usage: 25%          Users logged in: 0
Swap usage:   0%           IPv4 address for ens5: 10.0.2.155

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

ubuntu@ip-10-0-2-155:~$
```

Explanation

- Connects to Private EC2 through Bastion Host.

Step 16: Transfer File to Private EC2

Securely transfer PRN file.

Command (Public EC2)

`scp -i lab.pem prn.txt ubuntu@<PRIVATE_EC2_PRIVATE_IP>:/home/ubuntu/`

```
ubuntu@ip-10-0-2-155:~$ exit
logout
Connection to 10.0.2.155 closed.
ubuntu@ip-10-0-1-89:~$ scp -i del_20260607_lab_Vpc.pem prn.txt ubuntu@10.0.2.155:/home/ubuntu/
prn.txt
100% 18 26.0KB/s 00:00
ubuntu@ip-10-0-1-89:~$ ssh -i del_20260607_lab_Vpc.pem ubuntu@10.0.2.155
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1004-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sun Jun  7 17:32:12 UTC 2026

System load:  0.0           Temperature:   -273.1 C
Usage of /:   30.5% of 6.61GB Processes:      118
Memory usage: 26%          Users logged in: 0
Swap usage:   0%           IPv4 address for ens5: 10.0.2.155

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update
Failed to connect to https://changelogs.ubuntu.com/meta-release. Check your Internet connection or proxy settings

Last login: Sun Jun  7 17:30:16 2026 from 10.0.1.89
ubuntu@ip-10-0-2-155:~$ ls
prn.txt
ubuntu@ip-10-0-2-155:~$
```

Explanation

- Secure file transfer.

Step 17: Verify File on Private EC2

Confirm successful transfer.

Command

`cat prn.txt`

```
ubuntu@ip-10-0-2-155:~$ ls
prn.txt
ubuntu@ip-10-0-2-155:~$ cat prn.txt
PRN: 260240123060
ubuntu@ip-10-0-2-155:~$
```

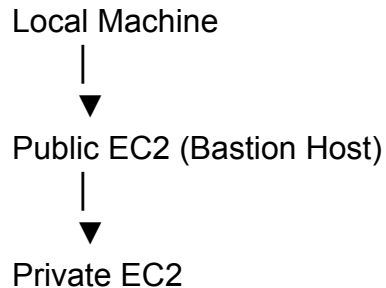
Expected Output

PRN: 260240123060

Step 18: Demonstrate Connectivity Flow

Verify complete Bastion Host architecture.

Connectivity Verification



Verification Evidence

1. SSH Local → Public EC2.
2. SSH Public EC2 → Private EC2.
3. File Transfer Public → Private.
4. File Verification on Private EC2.

Memory Sheet (Command Flow)

```
ssh -i lab.pem ubuntu@<PUBLIC_IP>
```

```
scp -i lab.pem lab.pem ubuntu@<PUBLIC_IP>:/home/ubuntu/
```

```
echo "PRN: 260240123060" > prn.txt
```

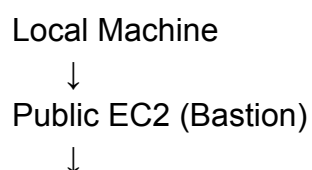
```
chmod 400 lab.pem
```

```
ssh -i lab.pem ubuntu@<PRIVATE_IP>
```

```
scp -i lab.pem prn.txt ubuntu@<PRIVATE_IP>:/home/ubuntu/
```

```
cat prn.txt
```

This flow demonstrates:



Private EC2



PRN File Transfer Verification

**PAISA NAHI HAI
APNE PAS SABKUCH
REVERSE ORDER
MAIN DELETE KARO!**